CIVE 202

Spring 2026

Project 1

Nebraska Air Quality Data Analysis

Submitted by

Elizabeth Wiggins

Submitted to

UNMC Water, Climate, and Health Group

February 3, 2026

Contents

Introduction.....3

Project Background 3
Scope of Work 3
Objectives 4
Method 4
Coding 4
Research 5
Results 5
References 8

Introduction

Project Background

UNMC started an initiative with AirPurple to collect data from air quality monitors distributed throughout Nebraska in an effort to better understand the impact that factors like water, air, and soil pollution have on human health. The extensive dataset that was produced for this initiative includes data recorded from February 23, 2024, to March 24, 2025, using sensors located throughout the entire state. The dataset includes columns for air humidity, pressure, temperature, volatile organic compounds, particulate matter, the altitude the sensor was placed at, and more.

Scope of Work

<u>Project Goals:</u>

The goal of this project is to provide an analysis and interpretation of an air quality dataset collected by monitors in various Nebraska locations for a request for proposal (RFP) from the UNMC Water, Climate, and Health Group. The client is looking for a breakdown of the impact that factors like temperature, humidity, and geographic location have on air quality in Nebraska. There are five specific parameters the client is interested in determining: (1) the Nebraska locations with the highest concentrations of volatile organic compounds (VOC) and particulate matter (PM) sizes 2.5 and 10.0, (2) the specific days these values were at their highest and an explanation for what causes them to be so high, (3) the effect humidity and temperature have on air quality, (4) the impact of the data sensor's altitude on the air quality values, and (5) if there were any air quality index (AQI) health risks in the sensor locations for PM 2.5 and PM 10.

<u>Project Tasks:</u>

There are two types of tasks to be carried out for this project.

The first task is using coding to manipulate and simplify the dataset for analysis. This will be done by using numerous commands such as filtering, sorting, grouping, and more.

The second task is research regarding the outcome of the data analysis. This research will make the analysis more comprehensive as it will explain what the numbers from the data indicate. Aspects to research consist of AQI values for the different PM sizes, which AQI's are considered unhealthy for sensitive groups, the causes of low air quality on the worst recorded days, and more as needed.

<u>Project Deliverables</u>

All files to be submitted by February 3, 2026, at 5:00pm CT

1. Raw data files used in the Python Code file
2. Python code file used to generate solution
3. Scope of work

4. Annotated code document (ACD)
5. Report

Objectives

The objective of the analysis is to simplify the dataset in a way that allows for conclusions to be drawn about the air quality in Nebraska. Specifically, how air quality relates to human health.

Method

Coding

To find the five locations in Nebraska with the highest mean and median concentrations of VOC, PM 2.5, and PM 10.0, begin by importing the pandas package and converting the data to a data frame. Then using a. groupby function, group the data by sensor name and aggregate the first of the columns, VOC, PM 2.5, or PM 10.0, by mean and assign it to a new variable. The data can then be assigned to another variable and sorted by mean with the condition that ascending values equals false, and only the head 5 values are to be included. This variable will then create a table of the 5 sensor locations with the largest values of whichever column chosen. This process can then be repeated but grouped and sorted by the median instead of the mean and listed as a new variable. This variable will create a table with the sensor locations that have the largest middle value of the given column, if the numerical range for values in that column were to be listed in order from largest to smallest. This process can be repeated for the remaining two columns to find their top five highest mean and median sensor locations in Nebraska.

The days with the highest recorded concentration of each column, VOC, PM 2.5, and PM 10.0, can be obtained by assigning a variable to the condition that the column value is greater than 0. Then assign another variable to a line that takes the previous variable, selects the date column, sensor name column, and the previously chosen column, and sorts this data by the previously chosen column. You can then take tail 1 to this variable to get the day with the highest concentration of the given column. This can be repeated for each of the other two variables to get their highest recorded day as well.

To simplify the temperature and allow for an easier comparison of patterns relating temperature to air quality, create a function that uses if, elif, and else conditions to set up a range for the different temperature levels such as, below freezing, cool, warm, and hot. Then the function can be applied to the temperature column of the dataset and printed into a table with the rest of the data. This can be repeated for the humidity levels to create a range that sorts the humidity percentage values into categories of low, high, and very high. Apply this function to the humidity column instead of the temperature.

Finally, the unhealthy PM events, for both 2.5 and 10.00, can be determined by condensing the dataset to values of PM greater than the value listed unsafe for sensitive groups by the AQI index and setting that equal to a new variable. Then assign another new variable to select the date,

sensor name, and PM columns, sorted by the PM values. This variable will list all the events of AQI health risks for that PM size in a table sorted into numerical order. Repeat procedure for the other particle size.

Research

Research included finding the locations each sensor was in and determining factors that could impact the air quality in that location. Particularly the locations with the highest recorded concentrations of VOC, PM 10.0, and PM5 were researched to learn what was causing poor air quality in those regions.

Additionally, AQI research was performed to understand what concentrations of PM constitute health risks for sensitive groups. These values were then utilized in the coding data analysis as well to sort the data only greater than these values into a table for comparison.

Results

The Nebraska cities with the highest recorded mean concentrations (listed from highest to lowest) of VOC are the following: Ogallala, York, Fremont, Norfolk, and McCook. Nebraska cities with the highest recorded median concentrations of VOC: Ogallala, McCook, Fremont, York, and Norfolk. The Nebraska cities with the highest recorded mean concentrations of PM 2.5: Broken Bow, Auburn, Tecumseh, O'Neill, and McCook. Nebraska cities with the highest recorded median concentrations of PM 2.5: Broken Bow, Auburn, Tecumseh, Norfolk, and Wisner. Nebraska cities with the highest recorded mean concentrations of PM 10: Broken Bow, Auburn, Tecumseh, O'Neill, and McCook. The Nebraska cities with the highest recorded median concentrations of PM 10: Broken Bow, Auburn, Norfolk, Tecumseh, and Wisner.

The maximum value of VOC concentration was recorded on June 24, 2024, in Ogallala. This high concentration of VOC is likely due to the extreme heat that day causing a lot of evaporation of their polluted groundwater. As shown in figure 1, the temperature in Ogallala this day was extremely high, with the AirPurple sensor (named Swnphd-ogallala) recording a temperature of 116º F. This high temperature in combination with the low humidity of 9%, as recorded by the sensor, made ideal conditions for evaporation. The groundwater is part of the Ogallala aquifer, which overlies "some of the most productive regions in the United States for ranching livestock, and growing corn, wheat, cotton, and soybeans" (Pradhan 2023) making it frequently polluted with pesticides, fertilizer, and various other pollutants in agricultural runoff. On days of extreme heat and low humidity, such as June 24, 2024, these pollutants in the water evaporate and fill the air with high levels of VOC. This demonstrates that humidity and temperature do have a noticeable effect on air quality, especially when combined with factors like water pollution.

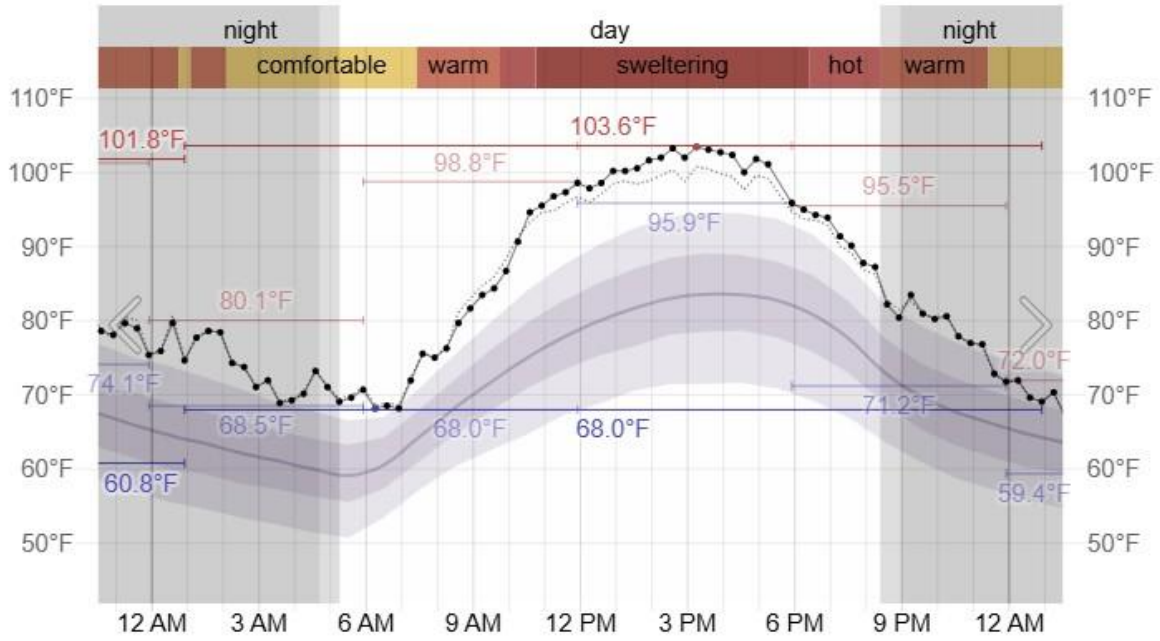

Figure 1 Weather chart of June 24, 2024, in Ogallala, NE (Weather Spark)

The maximum values of PM 2.5 and PM 10 concentrations were recorded on February 18, 2025, in Auburn. High PM concentrations are frequent in Auburn, NE, likely due to their large utilities industry. Because their utilities, gas, and mining industries are much larger than other areas of Nebraska, as demonstrated by figure 2, they also have a higher concentration of PM in their air since these industries are known to produce high levels of air pollution.

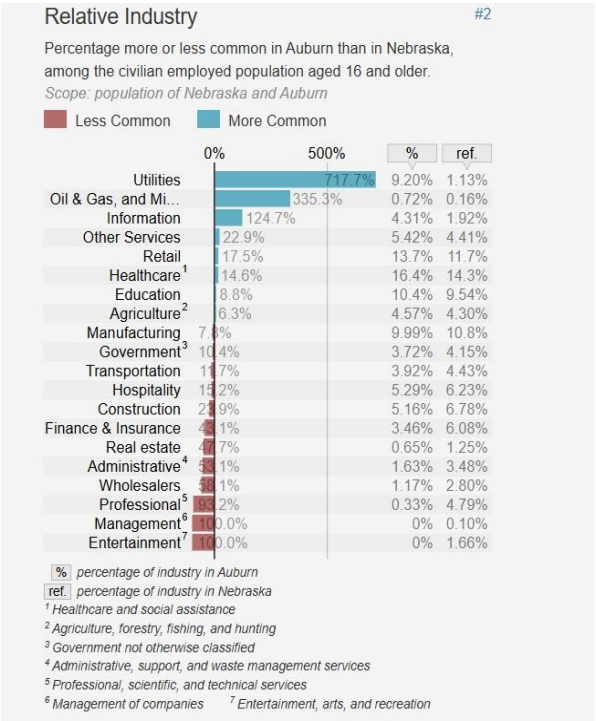

Figure 2 Relative Industry Graph (Statistical Atlas)

In the period that the sensor data was collected, February 23, 2024, to March 24, 2025, 391 AQI health risks occurred for PM 10, meaning that the air quality in the location was recorded was not safe for sensitive groups due to those particles. Additionally, there were 520 instances in this same period where the concentration of PM 2.5 was high enough to warrant AQI health risks. There are numerous causes contributing to these high concentrations including manufacturing and other industrial activities, agricultural pollution, smoke, and more. However, since there are so many instances of these health risks at various times of the year across the entire state, it makes narrowing down the exact causes difficult.

References

1.Industries in Auburn, Nebraska (city) https://statisticalatlas.com/place/Nebraska/Auburn/Industries#figure/relative-industry (accessed Feb 2, 2026).

2.June 24, 2024 weather history in Ogallala Nebraska, United States https://weatherspark.com/h/d/4782/2024/6/24/Historical-Weather-on-Monday-June-24-2024-inOgallala-Nebraska-United-States#Figures-ColorTemperature (accessed Feb 2, 2026).

3.Pradhan, D. The fate of Ogallala: A precious and Vulnerable resource https://clas.ucdenver.edu/ges/2023/05/04/fate-ogallala-precious-and-vulnerable-resource (accessed Feb 2, 2026).

4.Raw PM2.5 Air Quality Interactive map by PurpleAir https://map.purpleair.com/air-qualityraw-pm25?opt=%2F1%2Flp%2Fa10%2Fp604800%2FcC0#6.02/42.089/-97.537 (accessed Feb 2, 2026).

5.Technical Assistance Document for the Reporting of Daily Air Quality – the Air Quality Index (AQI) https://uofnebraskamy.sharepoint.com/personal/37381171_nebraska_edu/_layouts/15/onedrive.aspx?id=%2Fpersonal%2F37381171%5Fnebraska%5Fedu%2FDocuments%2FTeaching%2FCIVE%20202%2FSpring%202026%2FProject%20%231%2FEPA%20Air%20Quality%20Index%20Document%2Epdf&parent=%2Fpersonal%2F37381171%5Fnebraska%5Fedu%2FDocuments%2FTeaching%2FCIVE%20202%2FSpring%202026%2FProject%20%231&ga=1 (accessed Feb 2, 2026).